NeuroToxicology 56 (2016) 40–54

Contents lists available at ScienceDirect

NeuroToxicology

Full length article

The neuroprotective effect of lithium against high dose methylphenidate: Possible role of BDNF

Majid Motaghinejad[a], Zeinab Seyedjavadein[a], Manijeh Motevalian[a,*], Majid Asadi[b]

[a] Razi Drug Research Center & Department of Pharmacology, School of Medicine, Iran University of Medical Sciences, Tehran, Iran
[b] Department of Medical Biotechnology, School of Advanced Medical Technologies, Tehran University of Medical Sciences, Tehran, Iran

ARTICLE INFO

Article history:
Received 19 November 2015
Received in revised form 12 June 2016
Accepted 19 June 2016
Available online 22 June 2016

Keywords:
Methylphenidate
Lithium
Neurodegeneration
Anxiety
Depression
Cognition impairment
BDNF
Anti-oxidant
Inflammatory factors

ABSTRACT

Methylphenidate (MPH) is a neural stimulant with unclear neurochemical and behavioral effects. Lithium is a neuroprotective agent in use clinically for the management of manic-depressive and other neurodegenerative disorders. This study investigated the protective effect of lithium on MPH-induced oxidative stress, anxiety, depression and cognition impairment. Forty-eight adult male rats were divided randomly and equally into 6 groups. Treatment groups were received MPH (10 mg/kg) and various doses of lithium (75, 150 and 300 mg/kg) simultaneously and also lithium (150 mg/kg) alone for 21 days. Elevated Plus Maze and Forced Swim Test were used to determine the level of anxiety and depression in animals. Morris Water Maze was used to evaluate spatial learning and memory. The hippocampi of rats were isolated and the level and activity of oxidative, anti-oxidant and inflammatory factors were measured. Also brain derived neurotropic factor expression level was measured by RT-PCR and western blotting. MPH (10 mg/kg) caused behaviors indicative of anxiety and depression-like phenotypes in EPM and FST and cognition impairment in MWM. While lithium in all mentioned doses inhibited these effects. Treatment with MPH significantly increased lipid peroxidation, mitochondrial GSH content and IL-1β and TNF-α levels in isolated hippocampal cells. Moreover superoxide dismutase and glutathione peroxidase activities and BDNF expression remarkably decreased. Various doses of lithium attenuated these effects and significantly mitigated MPH-induced oxidative damage, inflammation and increased BDNF expression level. Lithium has the potential to act as a neuroprotective agent against MPH induced toxicity in rat brain and this might be mediated by BDNF expression in hippocampus of rats.

1. Introduction

Methylphenidate (MPH; Ritalin) is a neural stimulant that is normally prescribed for the management of attention deficit/hyperactivity disorder in children (Faraone et al., 2004; Greenhill et al., 2006). MPH binds to dopamine and norepinephrine reuptake pumps and inhibits the reuptake of these amines into the synaptic terminal, and significantly enhance their effects on receptors (Greenhill et al., 2006; Motaghinejad et al., 2015d). Due to pharmacological similarity to cocaine and amphetamines, it has high potential for abuse and addiction (Bruggisser et al., 2011). One of the important assumptions in using the animal models for study of the conditions related to human is that the employed model should reflect what happens in human subjects. To study the effects of methylphenidate on the brain and behavior, the bioavailability of MPH in the animal model reflects the human subjects. There is some experimental and clinical reports that tried to evaluate correlations between various doses of MPH and animal and human plasma levels (Balcioglu et al., 2009; Gerasimov et al., 2000; Hannestad et al., 2010; Quinn et al., 2004; Schiffer et al., 2006; Teo et al., 2002; Thanos et al., 2015; Volkow et al., 2001; Wargin et al., 1983). Some of these studies showed that MPH with doses of 0.75, 2.5, 3, 5, 10 and 20 mg/kg could provide plasma concentrations between 7.8–80.0 ng/ml. All these data suggested that there is a similarity between animal and human subjects in correlation of dose and plasma concentration of MPH (Balcioglu et al., 2009; Teo et al., 2002; Thanos et al., 2015; Wargin et al., 1983). Chronic administration of MPH can induce behavioral alterations such as anxiety and depression-like behavior in experimental animals (Motaghinejad et al., 2015c,d; Vendruscolo et al., 2008). Previous experiments have also demonstrated other

* Corresponding author at: Razi Drug Research Center & Department of Pharmacology, School of Medicine, Iran University of Medical Sciences, Hemmat high way, Beside Milad Tower, Postal code: 14496-14525,Tehran, Iran.
E-mail addresses: motevalian.m@iums.ac.ir, manijeh.motevalian@gmail.com (M. Motevalian).

http://dx.doi.org/10.1016/j.neuro.2016.06.010
0161-813X/

alterations such as cognition (learning and memory) impairment (Bedard and Tannock, 2007; Vendruscolo et al., 2008). Additionally, these studies confirmed that chronic abuse of MPH causes apoptosis and induces oxidative stress, which may damage the brain cells of young animals (Andreazza et al., 2007; Martins et al., 2006). MPH showed neurotoxic properties in some areas of the brain, such as the hippocampus (Fagundes et al., 2007). Methamphetamine-like agents may cause neurotoxicity in some areas of the brain (Motaghinejad et al., 2015e). Several studies have clearly shown that caspase protein family and DNA fragmentation are involved in methamphetamine-induced neurodegeneration (Andreazza et al., 2007; Riddle et al., 2006). Long term application of amphetamine-like agents dramatically activated neuroinflammation in brain cells and may increase cytokine expression (TNF-α, IL-1-β and IL-6) in rat central nervous system (Motaghinejad and Motevalian, 2016; Sadasivan et al., 2012). Previous studies confirm that acute and chronic abuse of methamphetamine stimulate the secretion of TNF-α and/or production of oxidative stress markers (Motaghinejad and Motevalian, 2016). Previous study have also demonstrated that the concentration of some antioxidant agents in brain tissues was decreased by administration of methamphetamine like compounds (Martins et al., 2006; Sandoval et al., 2002). Other studies indicated that chronic treatment of young rats with MPH presented a dose-dependent increase in TBARS content and protein carbonyls formation in specific rat brain regions (Martins et al., 2006; Schmitz et al., 2012). These studies also demonstrated that chronic exposure to the highest used dose of MPH increased lipid peroxidation in rat hippocampus (Schmitz et al., 2012). Lithium is used for the treatment of bipolar disorder, via the inhibition of glycogen synthase kinase-3β enzyme (GSK-3β) (Lenox and Hahn, 2000; Phiel and Klein, 2001). Many clinical and experimental studies showed that lithium from ranges 10–400 mg/kg caused neuroprotection in different conditions (Boyko et al., 2015; Chrislip et al., 1984; Dhawan et al., 1999; Frey et al., 2006; Hajek and Weiner, 2016; Khairova et al., 2012; Machado-Vieira et al., 2009; Rahimi et al., 2014; Rahimi_Balaei et al., 2010). Several clinical reviews showed that optimum lithium concentration after therapeutic doses is 0.6–0.75 mmol/l. However, the levels higher than 0.75 mmol/L may be involved in neuroprotection of this agent (Amdisen, 1967; Severus et al., 2007; Severus et al., 2008). On the other hand, some experimental studies in rat showed that pharmacokinetic pattern of lithium in rat is similar to human and neuroprotective doses produced plasma concentration of 0.3–0.7 mmol/l (Hanak et al., 2014; Morrison et al., 1971). Additionally, sub-chronic lithium treatment enhances the anxiolytic-like effects of serotonergic and noradrenergic drugs by inhibition of GSK-3β and facilitating central 5-HT and NE neurotransmission at therapeutic plasma level of lithium (Bauer et al., 2003; Mines et al., 2010; Tondo et al., 2014). Several studies have elucidated that lithium has neuroprotective properties (Chuang et al., 2011; Diniz et al., 2013b). Evidence from preclinical studies suggest that neuroprotection induced by lithium is mainly related to its potent inhibition of the enzyme GSK-3β (Jope, 2003; Yu et al., 2012). These results suggest that lithium has emerged as a robust neuroprotective agent in preventing apoptosis in neurons. This study also suggests that lithium causes activation of antioxidant enzymes such as GPx and SOD in rat hippocampus (Chuang, 2004; Chuang et al., 2002; Diniz et al., 2013a). Neuroprotective agents such as lithium can inhibit oxidative free radicals like MDA and lipid peroxidation and can diminish the level of some other oxidative agents such as GSSG (Khairova et al., 2012). Our recent studies showed that MPH in various doses caused neurodegeneration and increased inflammatory parameters and oxidative stress in an adult rat hippocampus. On the other hand, in recent years, chronic abuse of MPH in human adults have been markedly increased but the effect of MPH on adult rat brain, specially effect of chronic use, have not been fully clarified in rat hippocampus (Abraham et al., 2012; Challman and Lipsky, 2000; Motaghinejad et al., 2015e). Thus based on this concept and increasing frequency of MPH abuse by adults, it seems that prevention of neurodegenerative effects of MPH is a critical task in management of addiction. The aim of this study is to evaluate the possible role of pre-treatment with lithium on the modulation of MPH induced stress, anxiety, depression, cognition impairment, oxidative stress and inflammation.

2. Materials and methods

2.1. Animals

Forty-eight adult male Wistar rats (250–300 g, 10 weeks old) were provided for this study. They were randomly divided to 6 groups (8 in each) and housed in a controlled condition, temperature 22 ± 0.5 °C with 12-h light/dark cycles and had free access to food and water. This experimental protocol was set according to the Guide for the care and use of laboratory animals published by National Institute of Health (NIH) guidelines, publication No. 85-23, revised 1996.

2.2. Drugs

Methylphenidate (MPH) and Lithium were purchased from Sigma-Aldrich Co. All agents were freshly prepared just before use.

2.3. Experimental design

- Group 1 received normal saline (0.7 mL/rat, *i.p.*) for 28 days and served as control group.
- Group 2 received MPH (10 mg/kg, *i.p.*) alone for 28 days.
- Groups 3, 4 and 5 were treated with MPH (10 mg/kg, *i.p.*) and lithium (75, 150 and 300 mg/kg, *i.p.* respectively) simultaneously for 28 days.
- Group 6 received lithium (150 mg/kg, *i.p.*) alone for 28 days.

During the period of drug administration, in day 22, Elevated Plus Maze (EPM) was used for investigating the level of anxiety and in day 25, Forced Swim Test (FST) was used for assessment of depression.

In addition, between days 17 and 21, the Morris Water Maze (MWM) test was performed to evaluate the effect of MPH alone and MPH in combination with lithium on spatial learning and memory.

24 h after last drug administration and after evaluation of all behavioral signs (on day 28), rats were anaesthetized using thiopental (50 mg/kg, *i.p.*), their total hippocampi were isolated and various oxidative and anti-oxidant and inflammatory biomarkers were measured. In addition, according to our previous studies and others' work (Leutgeb et al., 2007; Motaghinejad et al., 2015f), two parts of hippocampus were isolated, dentate gyros (DG) and CA1, and BDNF expression in gene and protein levels were measured in these two areas. Histological studies were undertaken by crystal violet staining for the assessment of cell numbers and morphological changes in DG and CA1 areas of hippocampus.

The selection of doses of MPH was done according to previous work results on neurodegenerative doses of MPH in animal models and human subjects (Balcioglu et al., 2009; Gerasimov et al., 2000; Hannestad et al., 2010; Motaghinejad et al., 2015e; Quinn et al., 2004; Schiffer et al. 2006; Teo et al., 2002; Thanos et al., 2015; Volkow et al., 2001; Wargin et al., 1983), and neuroprotective doses of lithium was obtained from previous work on human and animal studies (Boyko et al., 2015; Chrislip et al., 1984; Dhawan et al., 1999; Frey et al., 2006; Hajek and Weiner, 2016; Khairova et al., 2012;

Machado-Vieira et al., 2009; Rahimi et al., 2014; Rahimi_Balaei et al., 2010). There are several references in relation to clinical relevance of choosing MPH and lithium doses in our experiment, and also the relationship that exist between doses and plasma levels in experimental subjects (Balcioglu et al., 2009; Hanak et al., 2014; Morrison et al., 1971; Severus et al., 2007, 2008; Teo et al., 2002; Thanos et al., 2015; Wargin et al., 1983).

2.4. Behavioral tests

2.4.1. Forced swim test (FST)

The Forced Swim Test (FST) is used for the evaluation of depressive-like behavior in rodents. The apparatus is composed of a transparent plexiglass cylinder with 30 cm diameter and 65 cm height; this was filled with water up to 30 cm. The day before the experiment, in order to have adaptation of animals, all animals were individually located to swim for a period of 15 min, on the day of experiment; animals were positioned individually for a period of 6 min inside the water filled cylinder. Swimming duration was 6 min. Swimming is considered as a non-depressive behavior (Motaghinejad et al., 2015c,d).

2.4.2. Elevated plus maze (EPM)

Elevated Plus Maze (EPM) was used to assess anxiety in animal models. The apparatus, which took the shape of a plus sign, included two crossing arms 60×20 cm, joined with a central square (10×10 cm). Two of the four arms were without walls while the two other arms had 40 cm elevated walls. All parts of an apparatus were kept 50 cm height above the ground. All subjects were situated individually in the center of the maze in front of a closed arm. The time that each animal spent on the open arms over a 5-min span was recorded. Spending a greater duration of time in open arms is correlated with non-depressive behavior (Faraone et al., 2004; Greenhill et al., 2006; Motaghinejad et al., 2015c,d).

2.4.3. Morris water maze task (MWM)

MWM was composed of a circular black-colored water tank (150 cm in diameter and 85 cm in height), which was set up in the center of small room. The apparatus was divided into four quadrants (North, East, West and South). The tank was filled by water to the height of 50 cm. During the experiment, the operator stayed in the north-east section of the room. A platform disk with 12 cm diameter (made of invisible plexiglass) was inserted 1 cm beneath the surface of the water. In the first 4 days of the experiment, which is termed the training procedure, the platform was located in one of the four quarters. An automated infrared tracking system (CCTV B/W camera, SBC-300 (P), Samsung electronics Co., Ltd., Korea) recorded the position of the animal in the tank. The camera was mounted 2.3 m above the surface of the water (Motaghinejad, et al., 2015c,d).

A) Handling

Before the start of the experiment, rats were taken from the cage to the tank that was filled with water, room temperature (25 ± 2 °C) and they were guided by hands of operator toward the platform which were placed on south-west quarter of the tank.

B) Training procedure

Pictorial landmarks, such as a picture, a window and a door, were set up in an extra maze in the room for spatial cues so that the rats could find the position of the platform more quickly. The platform was positioned in the south-west quarter of MWM tank with 25 cm distance from the edge of the tank and 1 cm beneath the surface of water. Each rat participated in four trials a day. Each animal was tested randomly from four quarters (north, east, west and south). If the rats could find the platform within 60 s, the trial was automatically stopped by a computer. The reported data on following parameters are the mean value of 4 days' trials:

1. Escape latency, defined as an average of the time required to find the hidden platform across all training days.
2. Traveled distance, defined as an average of the distance traveled to the hidden platform.
3. Speed of swimming, defined as an average of swimming speed during training days.

On the fifth day, termed probe day, the platform was removed and rats were thrown into the water from one of the above-mentioned directions (East) and the percentage of animals that swam to the target quarter (South-West quarter) was recorded (Motaghinejad et al. 2015c,d).

2.5. Mitochondrial preparation

The skull was dissected and the brain tissue was detached for isolation of hippocampus, isolation of dentate gyros (DG) and CA1 areas of hippocampus.

The isolated tissues of the hippocampus were homogenized in cold homogenization buffer (20 mM 4-morpholinepropanesulfonic acid, pH 7.2, 250 mM sucrose, 10 mM $MgCl_2$, 0.05 mM EGTA) using a homogenizer. Homogenate tissues were centrifuged at 400*g* for 10 min and the supernatant obtained was re-centrifuged at 12,000*g* for 15 min. The sediments were suspended again two more times in homogenization buffer and centrifuged at 12,000*g* for 10 min. Finally, the remaining sediments were mixed with homogenization buffer and stored in the refrigerator (4 °C). Total concentration of mitochondrial protein was determined by Bradford method. The homogenized cell solutions were analyzed for measurement of oxidative stress and apoptosis factors (Motaghinejad et al., 2015a,b; Motaghinejad and Motevalian, 2016).

2.5.1. Determination of lipid peroxidation

The amount of total lipid peroxidation was measured by evaluating the amount of thiobarbituric acid reactive compounds (TBARS). The quantity of this compound was calculated using an extension coefficient of 165 mM^{-1} cm^{-1} at 530 nm. Results were expressed as nmol/mg protein (Motaghinejad et al., 2015a,b; Serteser et al., 2002).

2.5.2. Determination of GSH and GSSG content

The reduced and oxidized forms of glutathione (GSH and GSSG, respectively) were measured using two different kits: Bioxytech GPx-340 and Bioxytech GR-340 (Oxis Research, USA) and the results were reported as nmol/mg protein (Capela et al., 2007; Motaghinejad et al., 2015a,b).

2.5.3. Determination of manganese superoxide dismutase activity

Activity of manganese superoxide dismutase (Mn-SOD) was evaluated using a special kit (Cayman Chemical Company, USA). The absorbance was detected at 450 nm and the findings were expressed as U/ml/mg protein (Motaghinejad et al., 2015a,b).

2.5.4. Determination of glutathione peroxidase (GPx) activity

The activity of GPx was measured by Bioxytec GPx-340 kits (Oxis Research, USA). Results were reported as mU/mg protein (Hsieh et al., 2015; Motaghinejad et al., 2015a,b).

2.5.5. IL-1β and TNF-α measurement

Concentrations of IL-1β and TNF-α in supernatant of hippocampal cells were measured using a commercially available ELISA kit (Genzyme Diagnostics, Cambridge, U.S.A.). Briefly, 96 well microtitre plates (Nunc) were coated with sheep anti-rat IL-1β and TNF-α polyclonal antibodies (2 mg/ml in bicarbonate coating buffer; 0.1 M NaHCO3, 0.1 M NaCl, pH 8.2, for 20 h at 48 °C), then washed three times with washing buffer (0.5 M NaCl, 2.5 mM NaH_2PO_4, 7.5 mM Na_2HPO_4, 0.1% Tween 20, pH 7.2). 100 ml of a 1% (w/v) ovalbumin (Sigma Chemical Co., Poole, Dorset, UK) solution in bicarbonate as coating buffer was added to each well and incubated at 37 °C for 1 h. After three washes, 100 ml of samples and standards were added and plates were incubated at 48 °C for 20 h. After three washes, 100 ml of the biotinylated sheep anti-rat IL-1β or TNF-α antibody (1:1000 dilutions in washing buffer containing 1% sheep serum, Sigma Chemical Co., Poole, and Dorset, UK) was added to each well. The further incubation was carried out for 1 h at room temperature. After three washes, 100 ml avidin-HRP (Dako Ltd., UK) (1:5000 dilution in wash buffer) was added to each well and plates were incubated at room temperature for 15 min. After three washes, 100 ml of 3, 3′, 5, 5′-Tetramethylbenzidine (TMB) substrate solution (Dako Ltd., UK) was added to each well and the plates were incubated for 10 min at room temperature. At the end of the incubation period, 100 ml of 1 M H_2SO_4 was added to each well to stop the reaction and to facilitate the color development. Absorbance was read at 450 nm on a microtitre plate reader. The detection limit of the assay was determined to be 4.3 pg/ml. Results were expressed as ng IL-1β/ml or TNF-α/ml (Ji et al., 2015; Kuloglu et al., 2002; Motaghinejad et al., 2015b).

2.6. Real-time reverse transcriptase-PCR (RT-PCR) studies

Total RNA was extracted from ~200 μg of each area of hippocampus, DG and CA1, using ONE STEP-RNA reagent (Bio Basic, Canada inc.) according to the manufacturer's instruction and the quantity and quality of RNA were analyzed using a nanodrop (ND-1000, Thermo Scientific Fisher, US) and gel electrophoresis. To eliminate any genomic contamination, RNA was treated with DNase I (Qiagen, Hilden, Germany) as described by the manufacturer. Complementary DNA (cDNA) was synthesized using 1 μg of total RNA. The integrity and quality of cDNA were examined with GAPDH primers as housekeeping. Real-time reverse transcriptase-PCR (RT-PCR) was carried out to evaluate the differences in expression patterns of BDNF gene among samples of each group. The specific primers corresponding to the coding sequence including BDNF Forward: 5′-GGAGGCTAAGTGGAGCTGAC-3′; BDNF Reverse: 5′-GCTTCCGAGCCTTCCTTTAG-3′; GAPDH forward: 5′-AGACAGCCGCATCTTCTTGT-3′; GAPDH Reverse: 5′-CCGTTCACACC-GACCTTCA-3′ were designed for BDNF and GAPDH respectively by Primer 3 software version 0.4 (frodo.wi.mit.edu). Real time RT-PCR was performed in 20 μl reactions containing 1 μl cDNA target, 100 nM forward and reverse primers and 1 × SYBR® Premix Ex Taq™ II (Takara, Tokyo, Japan). Experiments were carried out in triplicate using a CFX96™ Real-Time System (C1000TM Thermal Cycler) (Bio-Rad, Hercules, CA, USA). Amplification conditions were as follow: initial denaturation at 95 °C for 10 min, followed by 40 cycles (denaturation at 95 °C for 15 s and annealing and extension at 60 °C for 60 s). The relative value of the mRNA expression level of BDNF gene was calculated by comparing the cycle thresholds (CTs) of the target gene with that of housekeeping gene (GAPDH) using the $2^{-\Delta\Delta ct}$ method and REST 2009 software. Serial dilutions of cDNAs were used for calculation of the efficiencies of the primer sets on real-time PCR. In this regard, it was found that the efficiencies of the various primer sets were similar (Motaghinejad et al., 2015f).

2.7. Western blotting

We studied the immunoreactivity BDNF contents of the DG and CA1 areas of an isolated hippocampus by western blotting. Electro transfer of the resolved bands from gel to polyvinylidene difluoride (PVDF) membrane (Millipore, Bedford, USA) was performed for 90 min at 0.7 mA/cm^2 by using a semi-dry transfer apparatus (PeQlab, VWR Company, Pennsylvania, USA). After transferring step, the membrane was weakly stained for about 3 min with Coomassie blue G-250 (Sigma Aldrich, UK) 1 μg/100 ml distilled water without methanol. Then the membrane was dried and cut into 2 mm wide stripes. After destaining with methanol, the strips washed and blocked with 2% BSA overnight at 4 °C and then added with 1:100 diluted human or ovine sera at room temperature (RT) for 2 h on a shaker. The membranes then were washed with PBS-T (three washing steps in a total of 10 min) and incubated with anti BDNF conjugated polyclonal anti-rabbit antibody (1:500 dilutions in BSA, 360 min, RT; Sigma Aldrich, Germany) and anti GAPDH, as housekeeping gene, conjugated polyclonal anti-rabbit antibody (1:1000 dilutions in BSA, 360 min, RT; Sigma Aldrich, Germany) or HRP conjugated polyclonal rabbit anti-sheep antibody (1:5000 dilution in BSA,120 min, RT; Sina Biotech, Iran) respectively. The strips were washed and incubated with chemiluminescence substrate (Luminol and H_2O_2) for 2 min in RT. Finally, the reactive bands detected on X-ray film within 10–20 s under safelight condition and density of band was calculated by comparing the target gene with that of housekeeping gene by Alpha ease soft ware (version 3.1.2, Alpha, Alpha Innotech Corporation, San Lernadro, United states) and reported as relative expression of BDNF.

2.8. Histological studies

The 10 μm thick coronal sections were serially collected from the left hemisphere. After staining with crystal violet, images of 20 selected coronal sections were captured at 400 × magnifications and were analyzed by the morphometric software (Optikavision pro, Italy). The cells quality, integrity and shape were determined in an area of 1, 300,000 μm^2 from hippocampal subfields in all sections.

2.9. Statistical analysis

The data were analyzed by GraphPad PRISM v.6 Software and averaged in every experimental group and expressed as means ± SEM. The differences between control and treatment groups were evaluated by ANOVA. To evaluate the severity of behaviors, the differences between means in groups were compared using the Tukey's test at a significant level of $P < 0.05$.

3. Results

3.1. Results of the forced swim test (FST)

Compared to the control group, animals in the MPH-treated group (10 mg/kg) displayed shorter FST swimming times with $F(5, 42) = 18.65$ and $p < 0.05$ (Fig. 1), Although lithium in all doses inhibited the effects of MPH and increased swimming time, just in dose of 300 mg/kg can increase the time of swimming prominently compared to the group receiving MPH (10 mg/kg) alone, with $F(5, 42) = 18.65$ and $p < 0.05$ (Fig. 1). Animals treated with lithium (150 mg/kg) alone displayed longer swimming time in FST which this alteration was significant in comparison to group under treatment by MPH (10 mg/kg) alone with $F(5, 42) = 18.65$ and $p < 0.05$ and was not significant compared to the control group (Fig. 1).

3.2. Results of elevated plus maze (EPM)

Animals in the control group spent more time in the open arms of the maze in comparison to the group treated by MPH (10 mg/kg) with $F(5, 42) = 9.12$ and $p < 0.05$ (Fig. 2). The result of our study indicated that animals treated with 150 and 300 mg/kg of lithium were more likely to spend a longer period of time in the open arm

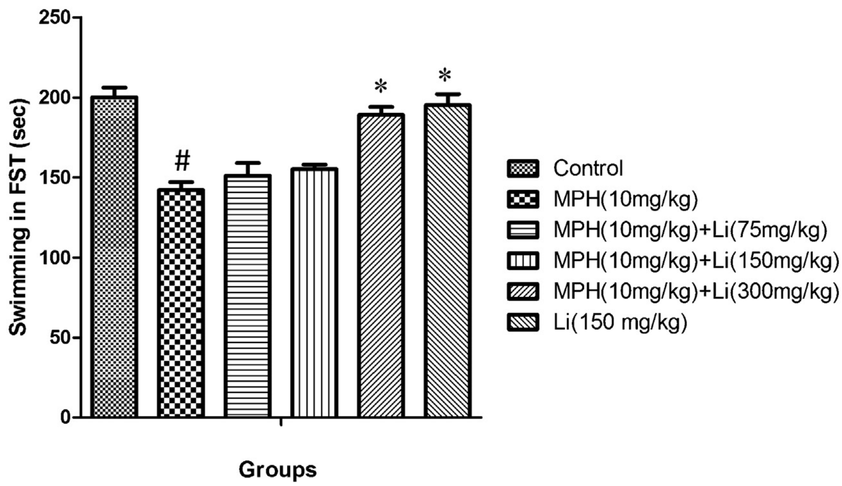

Fig. 1. Swimming Time (seconds) in control group, Lithium treated group, and groups under treatment with 10 mg/kg of MPH and the same doses of MPH in combination with Lithium by doses of 75, 150 and 300 mg/kg. All data are expressed as Mean ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

of the maze with F (5, 42) = 9.12 and $p < 0.05$ in comparison to animals treated with MPH (10 mg/kg) alone (Fig. 2). Animals treated with lithium (150 mg/kg) alone significantly spent more time in the open arms of the maze in comparison to group treated by MPH (10 mg/kg) alone with F (5, 42) = 9.12 and $p < 0.05$, while this change was not significant in comparison to control group (Fig. 2).

3.3. Evaluation of escape latency and traveled distance during training days in MWM

Average of escaped latency, with F (5, 42) = 5.72, and traveled distance, with F (5, 42) = 3.29, during the four days training period in the MWM for the MPH (10 mg/kg) treated group was statistically significant as compared to the control group ($P < 0.05$) (Figs. 3 and 4). Lithium in all doses inhibited MPH-induced increase in escape latency, with F (5, 42) = 5.72, and traveled distance, with F (5, 42) = 3.29, compared to the group treated with MPH (10 mg/kg) alone (Figs. 3 and 4). Animals treated with lithium (150 mg/kg) alone significantly showed higher average of escaped latency and traveled distance in comparison to group treated by MPH (10 mg/kg) alone with F(5, 42) = 5.72, and $p < 0.05$, while this change was not significant compared to the control group (Figs. 3 and 4).

3.4. Evaluation of swimming speed during training days

The swimming speed, with F(5, 42) = 0.97, was not altered during training trials in any of the animal groups, suggesting that exposure to lithium (150 mg/kg) alone, MPH (10 mg/kg) alone or in combination with lithium with all doses did not cause any motor disturbances (Fig. 5).

3.5. Evaluation of percentage of presence in target quarter in probe trial

Our data demonstrates that, relative to the control group, there was a decrease in the presence of animals in the target quarter in the MPH-treated group (10 mg/kg), with F(5, 42) = 11.90 and $P< 0.05$ (Fig. 7). Lithium in all doses dampened these effects of MPH (10 mg/kg) with F (5, 42) = 11.90 and $P< 0.05$ (Fig. 6). Animals treated with lithium (150 mg/kg) alone significantly spent more time in the target quarter of MWM in comparison to group treated

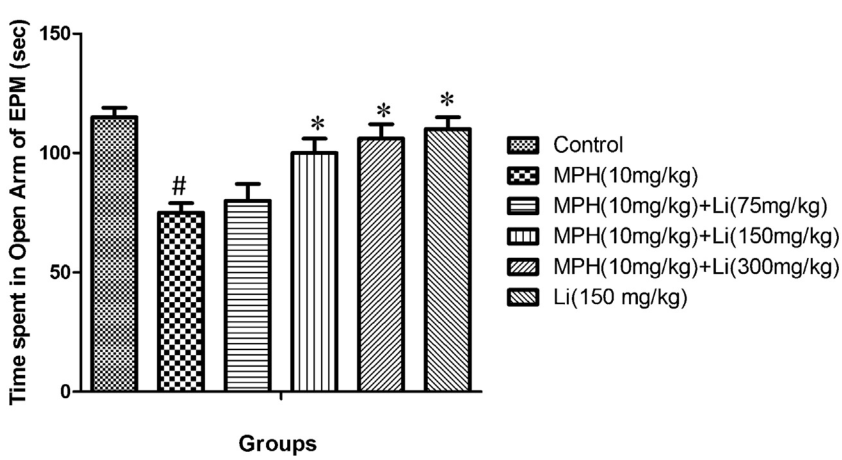

Fig. 2. Duration of time spent in open arms (seconds) in Elevated Plus Maze (EPM) test in control group, Lithium treated group and groups under treatment with 10 mg/kg of MPH and the same doses of MPH in combination with Lithium by doses of 75, 150 and 300 mg/kg. All data are expressed as Mean ± SD, (N = 8). * $P < 0.05$ vs 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

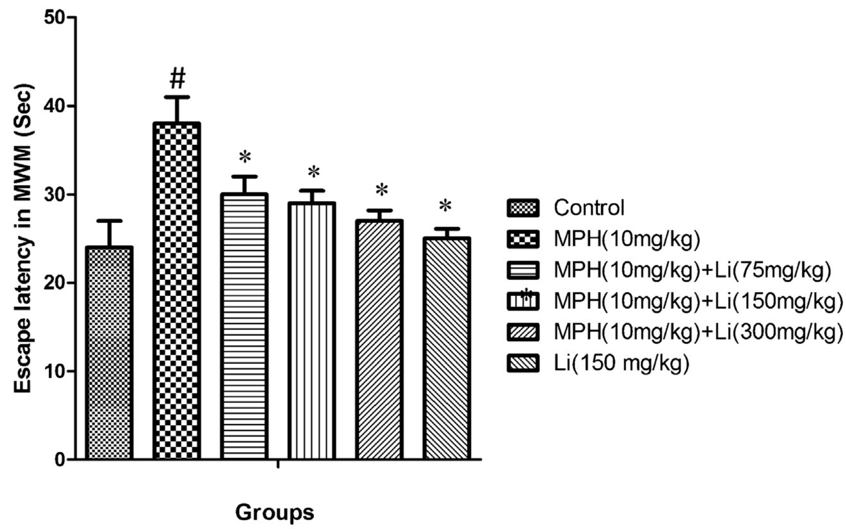

Fig. 3. Average of escaped latency in control group, Lithium treated group and groups under treatment with 10 mg/kg of MPH and the same doses of MPH in combination with Lithium by doses of 75, 150 and 300 mg/kg across all training days using Morris Water Maze (MWM) in rats. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

by MPH (10 mg/kg) alone with F (5, 42) = 11.90 and $p < 0.05$, while this change was not significant compared to the control group (Fig. 6).

3.6. MDA levels in isolated mitochondria

Treatment with MPH caused a statistically significant increase in lipid peroxidation and MDA levels in isolated mitochondria as compared to the control group with F(5, 42) = 9.07 and $P < 0.05$. Various doses of lithium (especially in 150 & 300 mg/kg) attenuated the MDA concentrations and this decrease had a prominent role compared to group treated by MPH (10 mg/kg) with F (5, 42) = 9.07 and $P < 0.05$ (Fig. 7). Treatment of animals by lithium (150 mg/kg) alone significantly decreased MDA level in comparison to group treated by MPH (10 mg/kg) alone with F (5, 42) = 9.07 and $p < 0.05$, while this alteration was not significant compared to the control group (Fig. 7).

3.7. GSH and GSSG levels in isolated mitochondria

As compared to the control group, treatment with MPH alone caused a significant decline in GSH content of mitochondria with F (5, 42) = 7.82 and $P < 0.05$. As compared to treatment with MPH-alone, treatment with MPH (10 mg/kg) and various doses of lithium (75,150 and 300 mg/kg) simultaneously increased the amount of isolated GSH significantly with F (5, 42) = 7.82 and $P < 0.05$ (Table 1). GSSG level in MPH treated group was significantly higher than those in control group with F (5, 42) = 0.17 and $P < 0.05$. Lithium (75, 150 and 300 mg/kg) prevented the increase of GSSG level induced by MPH and this decrease was significant in comparison to group under treatment by MPH (10 mg/kg) with (5, 42) = 0.17 and $P < 0.05$ (Table 1). Treatment of animals by lithium (150 mg/kg) alone significantly increased GSH level and decreased GSSG level in comparison to group treated by MPH (10 mg/kg) alone with F(5, 42) = 0.17 and $p < 0.05$, while it was not significant in comparison to control group (Table 1).

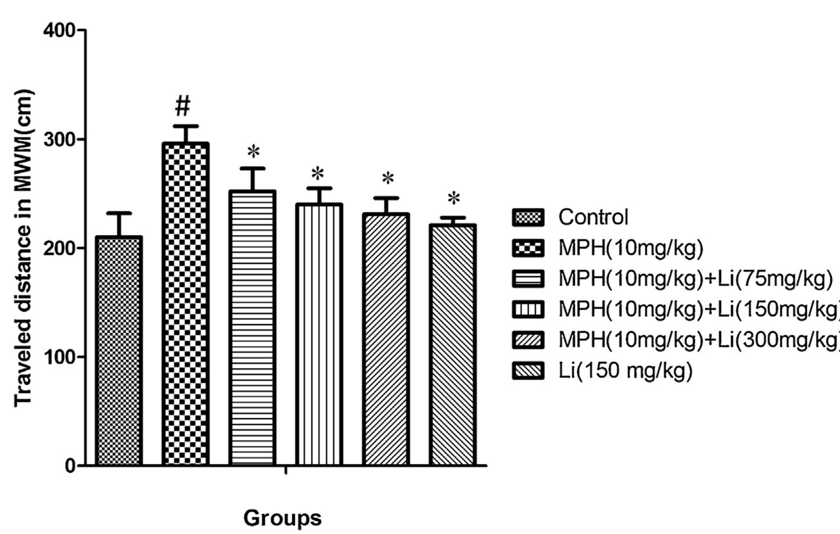

Fig. 4. Average of traveled distance in control group, Lithium treated group and groups under treatment with 10 mg/kg of MPH and the same doses of MPH in combination with Lithium by doses of 75, 150 and 300 mg/kg across all training days using Morris Water Maze (MWM) in rats. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

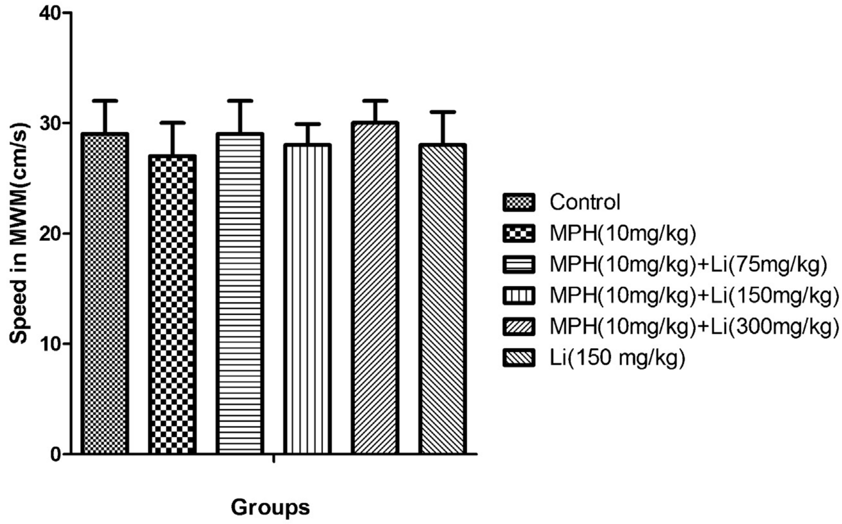

Fig. 5. Average of swimming speed in control group, Lithium treated group and groups under treatment with 10 mg/kg of MPH and the same doses of MPH in combination with Lithium by doses of 75, 150 and 300 mg/kg across all training days using Morris Water Maze (MWM) in rats. Data are shown as means ± SD, (N = 8), MPH: Methylphenidate. Li; Lithium.

3.8. Superoxide dismutase (SOD) activity in isolated mitochondria

Administration of 10 mg/kg of MPH caused significant inhibition of superoxide dismutase (SOD) activity with F (5, 42) = 7.44 and P < 0.05 relative to the control group. Additionally, administration of various doses of lithium (75, 150 and 300 mg/kg) significantly increased the activity of manganese superoxide dismutase when compared to the group that received MPH only with F (5, 42) = 7.44 and P < 0.05 (Fig. 8). Treatment of animals by lithium (150 mg/kg) alone significantly increased SOD activity in comparison to group treated by MPH (10 mg/kg) alone with F (5, 42) = 7.44 and p < 0.05, while it was not significant compared to the control group (Fig. 8).

3.9. Glutathione peroxidase (GPx) activity in isolated mitochondria

MPH (10 mg/kg) caused attenuation of glutathione peroxidase (GPx) activity and significantly decreased its activity relative to the control group with F (5, 42) = 18.60 and P < 0.05. Injection of various doses of lithium (75, 150 and 300 mg/kg) in rats treated with MPH increased glutathione peroxidase activity (GPx), with statistically significant increases observed with administration of 150 mg/kg and 300 mg/kg of lithium with F (5, 42) = 18.60 and P < 0.05 (Fig. 9). Treatment of animals by lithium (150 mg/kg) alone significantly increased GPx activity in comparison to group treated by MPH (10 mg/kg) alone F (5, 42) = 18.60 and p < 0.05, while it was not significant in comparison to the control group (Fig. 9).

3.10. Effects of various doses of lithium on MPH induced IL-1β and TNF-α level

MPH induced increases in IL-1β, with F (5, 42) = 25.09, and TNF-α level, with F (5, 42) = 21.26. These were significantly greater compared to the negative control group (P < 0.05). Lithium (150 and 300 mg/kg) prevented MPH induced inflammation and this decrease was statistically significant in comparison with MPH

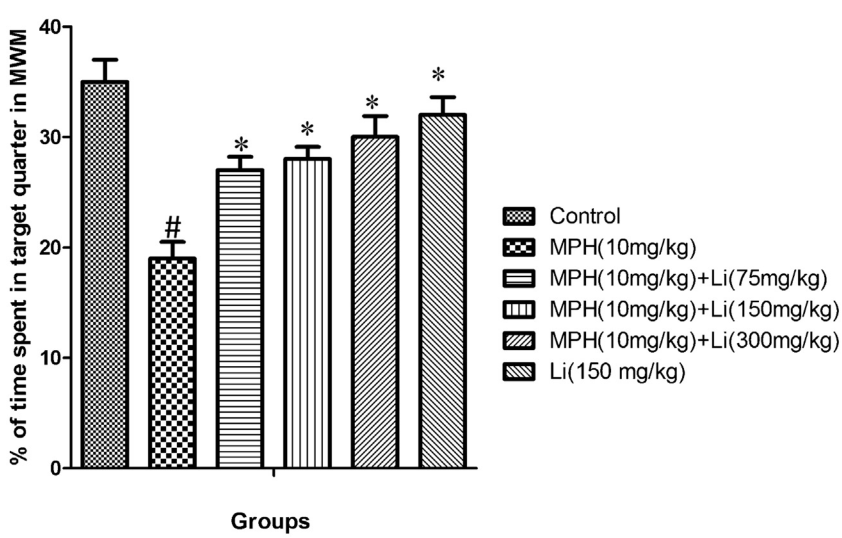

Fig. 6. Percentages of time spent in target quarter in probe trial in control group, Lithium treated group and groups under treatment with 10 mg/kg of MPH and the same doses of MPH with Lithium by doses of 75, 150 and 300 mg/kg across probe day in Morris Water Maze (MWM) in rats. Data are shown as means ± SD, (N = 8).* P < 0.05 vs 10 mg/kg of MPH. # P < 0.05 vs control group. MPH: Methylphenidate. Li; Lithium.

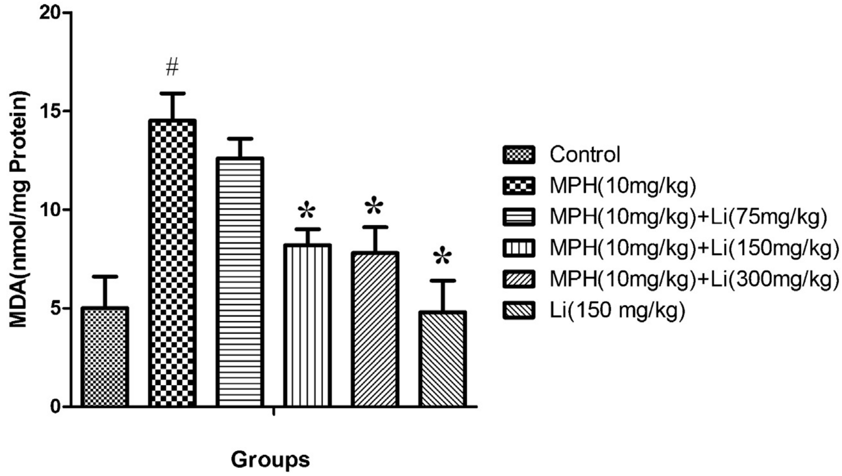

Fig. 7. Effects of MPH (10 mg/kg), lithium (150 mg/kg) and various doses of Lithium (75, 150 and 300 mg/kg) on MPH-induced lipid peroxidation in rat isolated hippocampus mitochondria. Data are shown as means ± SD, (N = 8).* P < 0.05 vs. 10 mg/kg of MPH. # P < 0.05 vs. control group. MPH: Methylphenidate. Li; Lithium.

Table 1
Effects of various doses of Lithium on methylphenidate induced GSH/GSSG alterations in mitochondria.

Group	GSH (nmol/mg protein)	GSSG (nmol/mg protein)	GSH/GSSG
Control	60.2 ± 4.1	0.78 ± 0.1	76
MPH (10 mg/kg)	42.6 ± 2.1 [a]	3.3 ± 1.6 [a]	12.72 [a]
MPH (10 mg/kg) +Li (75 mg/kg)	46.4 ± 1.4 [b]	2 ± 0.05 [b]	23 [b]
MPH (10 mg/kg) +Li (150 mg/kg)	48.4 ± 2 [b]	1.93 ± 0.13 [b]	25.26 [b]
MPH (10 mg/kg) +Li (300 mg/kg)	53.5 ± 2.9 [b]	1.72 ± 0.16 [b]	31.1 [b]
Li (150 mg/kg)	58.6 ± 1.4 [b]	0.92 ± 0.2 [b]	64 [b]

All data are given as Mean ± SEM, (N = 8).
[a] Shows significant difference from negative control group (p < 0.05).
[b] Shows significant difference from the group treated with methylphenidate (p < 0.05).MPH: methylphenidate; Li: lithium

(10 mg/kg) only group, with F (5, 42) = 21.26 for TNF-α level and with F (5, 42) = 25.09 for IL-1β, (P < 0.05) (Figs. 10 and 11). Treatment of animals by lithium (150 mg/kg) alone significantly decreased IL-1β, with F (5, 42) = 25.09, and TNF-α level, with F (5, 42) = 21.26, in comparison to group under treatment with MPH (10 mg/kg) alone (p < 0.05); while it was not significant in comparison to the control group (Figs. 10 and 11).

3.11. Effects of various doses of lithium on MPH induced decrease in expression of BDNF gene

MPH induced decrease in expression of BDNF gene in both DG, with F (5, 42) = 3.14, and CA1, with F (5, 42) = 2.64, regions of hippocampus in comparison to the negative control group

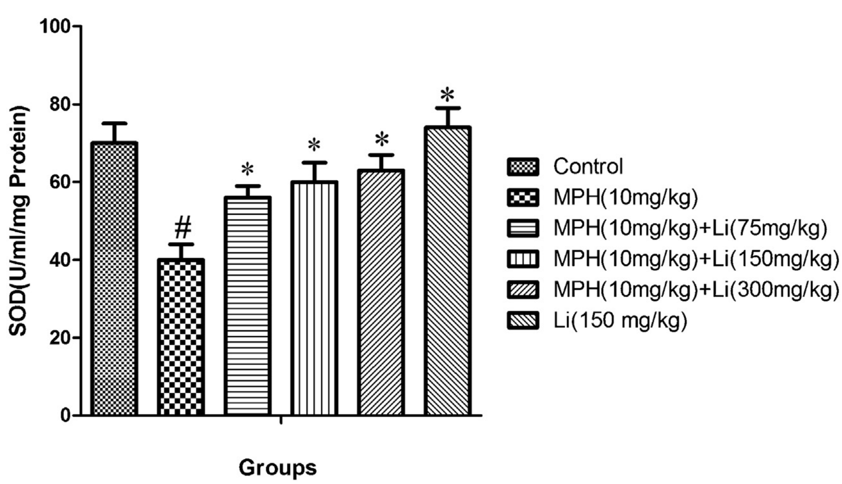

Fig. 8. Effects of MPH (10 mg/kg), lithium (150 mg/kg) and various doses of Lithium (75, 150 and 300 mg/kg) on MPH-induced alterations in manganese superoxide dismutase (SOD) activity in rat isolated hippocampus mitochondria. Data are shown as means ± SD, (N = 8).* P < 0.05 vs. 10 mg/kg of MPH. # P < 0.05 vs. control group. MPH: Methylphenidate. Li; Lithium.

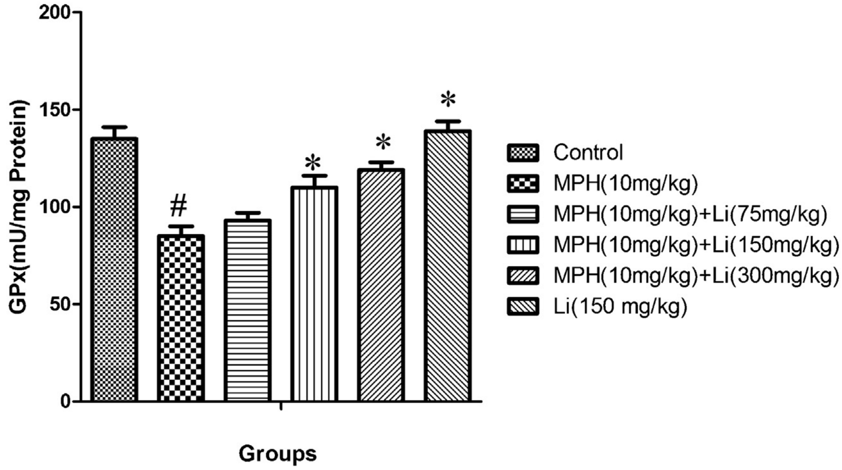

Fig. 9. Effects of MPH (10 mg/kg), lithium (150 mg/kg) and various doses of Lithium (75, 150 and 300 mg/kg) on MPH-induced alterations in glutathione peroxidase (GPx) activity in rat isolated hippocampus mitochondria. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs 10 mg/kg of MPH. # $P < 0.05$ vs control group. MPH: Methylphenidate. Li; Lithium.

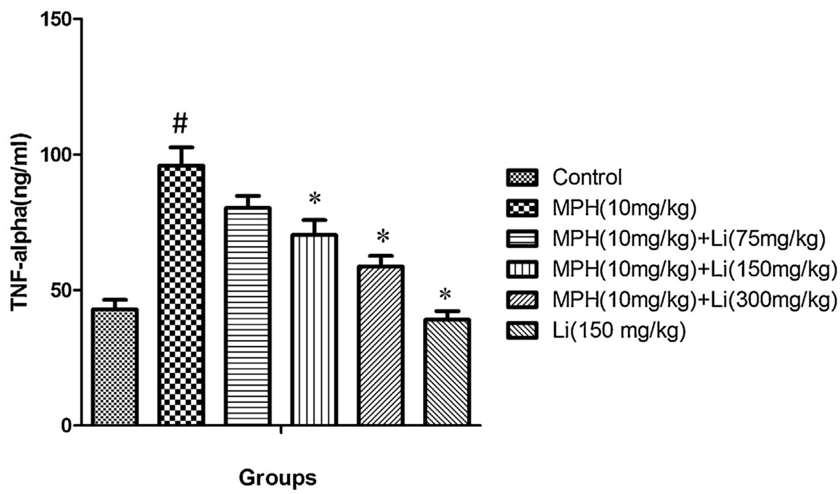

Fig. 10. Effects of MPH(10 mg/kg), lithium(150 mg/kg) and lithium (75, 150 and 300 mg/kg) on MPH induced TNF-α alterations in rat isolated hippocampus mitochondria. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

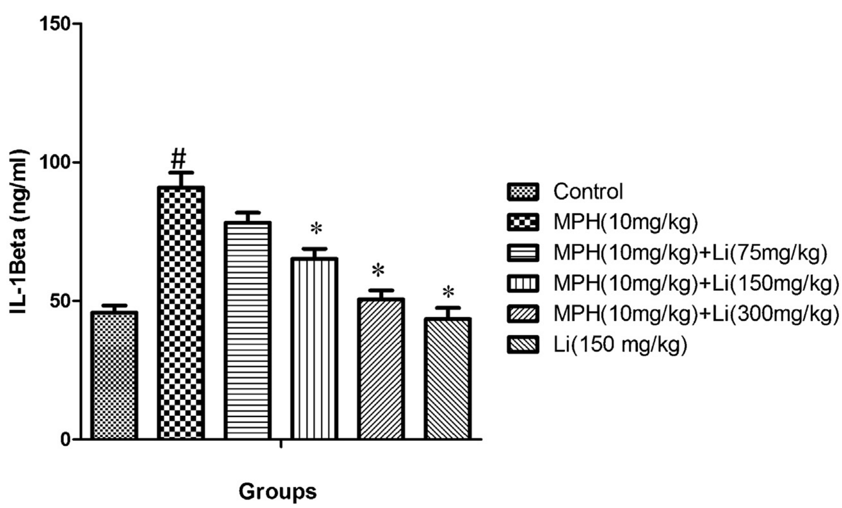

Fig. 11. Effects of MPH (10 mg/kg), lithium (150 mg/kg) and lithium (75, 150 and 300 mg/kg) on MPH induced IL-β alterations in rat isolated hippocampus mitochondria. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

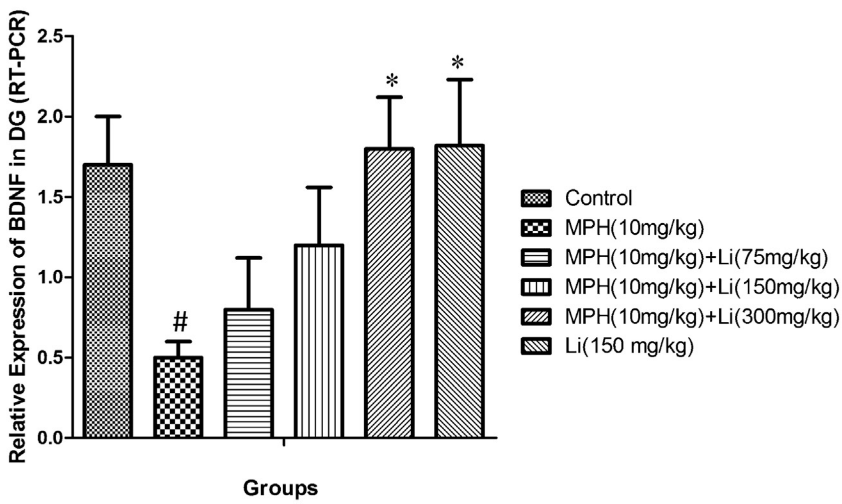

Fig. 12. Alterations of expression (RT-PCR) of BDNF in the control group and group under treatment with lithium (150 mg/kg) and 10 mg/kg of MPH and MPH combination with 75, 150 and 300 mg/kg of lithium in DG region of hippocampus. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

($P < 0.05$). Groups under treatment by lithium (150 mg/kg) alone and Lithium (300 mg/kg) +MPH cause increase of BDNF gene expression in DG, with $F(5, 42) = 3.14$, and CA1, with $F(5, 42) = 2.64$, region of hippocampus in comparison to the MPH (10 mg/kg) alone ($P < 0.05$) (Figs. 12 and 13).

3.12. Effects of various doses of lithium on MPH induced decrease in BDNF protein production

MPH induced decrease in BDNF protein production in both DG, with $F(5, 42) = 3.47$, and CA1, with $F(5, 42) = 2.73$, regions of hippocampus in comparison to the negative control group ($P < 0.05$). Groups under treatment by Lithium (150 mg/kg) alone and Lithium (300 mg/kg) + MPH cause increase in BDNF protein production in DG, with $F(5, 42) = 3.47$, and CA1, with $F(5, 42) = 2.73$, regions of hippocampus in comparison to the MPH (10 mg/kg) alone (Figs. 14 and 15). Treatment of animals by lithium (150 mg/kg) alone significantly increased BDNF protein production in both DG and CA1 regions of hippocampus in comparison to group treated by MPH (10 mg/kg) alone ($p < 0.05$) while it was not significant compared to the control group (Figs. 14 and 15).

3.13. Histological studies

MPH (10 mg/kg) caused dramatic degeneration in DG and pyramidal cells in CA1 regions in comparison to control group ($p < 0.05$), while lithium (150 and 300 mg/kg) caused inhibition of this effect of MPH and increased the number of cells in both regions ($p < 0.05$). The degenerated, shrunken dark cells with condensed nucleus were observed in granular and pyramidal cells in groups treated with 10 mg/kg of MPH. Lithium (150 and 300 mg/kg) inhibited the histological changes produced by MPH in these areas of hippocampus (Figs. 16 and 17).

4. Discussion

It has been observed in present study for the first time that treatment by various doses of lithium can attenuate chronic MPH

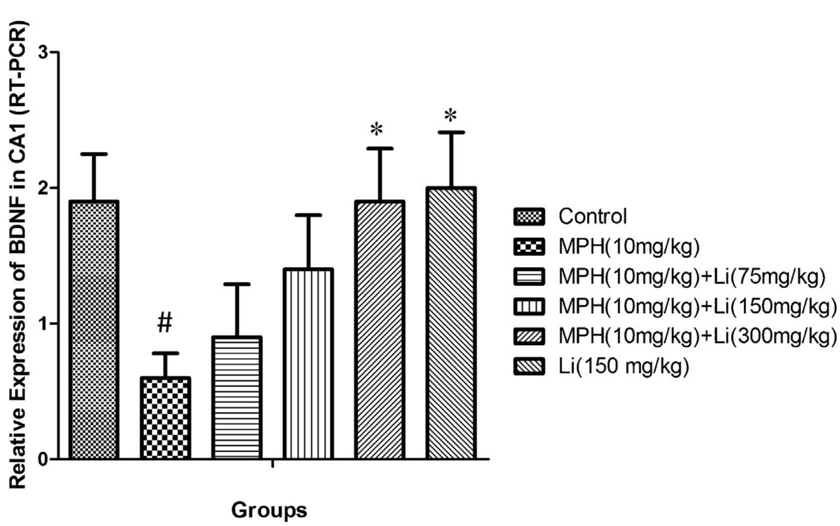

Fig. 13. Alterations of expression (RT-PCR) of BDNF in the control group and group under treatment with lithium (150 mg/kg) and 10 mg/kg of MPH and MPH combination with 75, 150 and 300 mg/kg of lithium in CA1 region of hippocampus. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

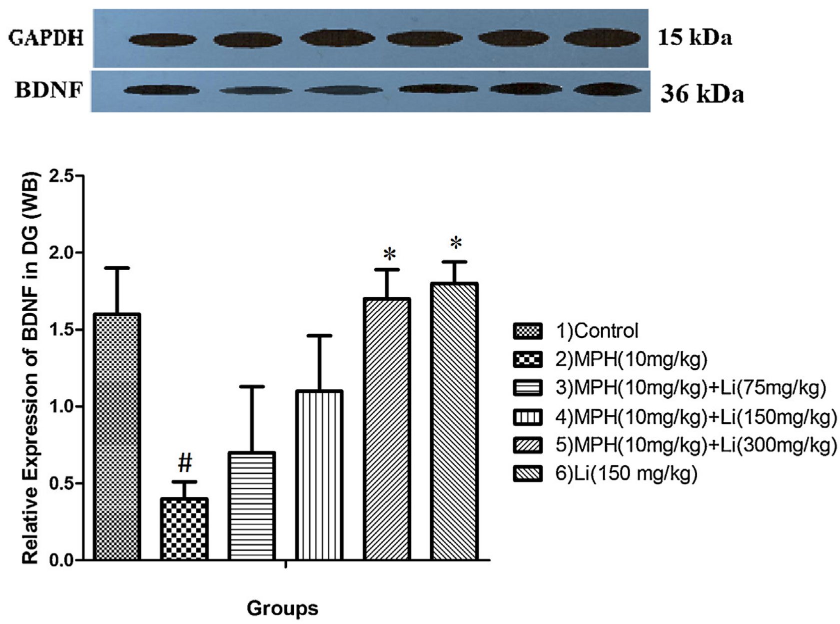

Fig. 14. Alterations of relative expression (Western blot) of BDNF in the control group, Lithium treated group and group under treatment with 10 mg/kg of MPH and its combination with 75, 150 and 300 mg/kg of lithium in DG region of hippocampus. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

induced neurodegeneration which is characterized by oxidative stress and inflammation in isolated hippocampus in adult rats. This study has demonstrated that lithium in a dose dependent manner can prevent MPH (10 mg/kg) induced anxiety-like behavior in EPM and depressive-like sign in FST and cognition impairment in MWM. Also lithium in a dose dependent manner can prevent MPH induced increase in lipid peroxidation, GSSG content, TNF-α and IL-1β, also prevent MPH induced decrease in GSH content, GPx and SOD activities in isolated rat hippocampus. Moreover lithium increases BDNF in gene and protein levels in isolated hippocampus of MPH treated rats and prevent MPH induced hippocampus cell degeneration. Our recent study showed the effect of various doses of MPH (2, 5, 10 and 20 mg/kg) on adult rat hippocampal cells (Motaghinejad et al., 2015e), and another study showed the effects of various doses of lithium in neuroprotection against some neurodegenerative disorders (Marmol, 2008). According to these

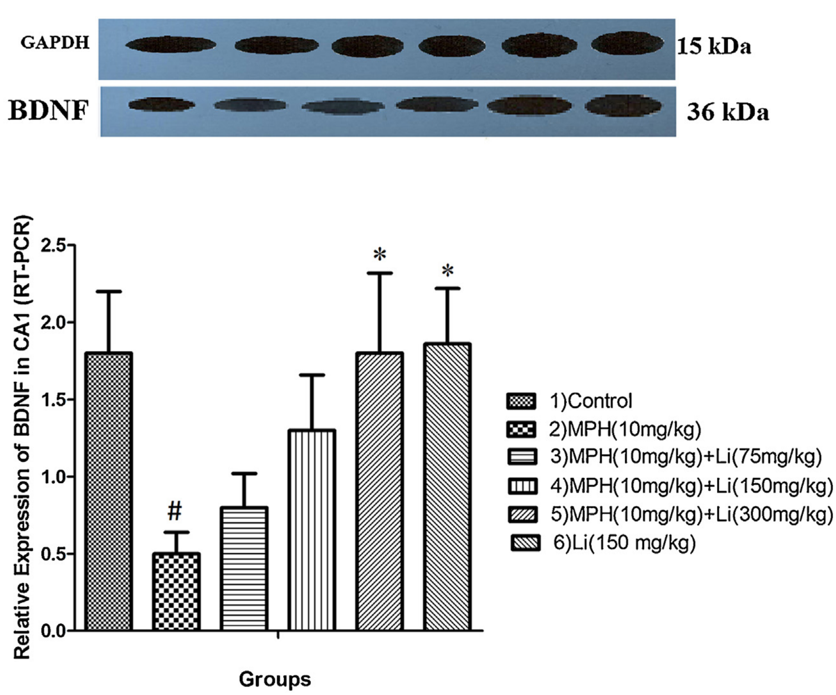

Fig. 15. Alterations of relative expression (Western blot) of BDNF in the control group, Lithium treated group and group under treatment with 10 mg/kg of MPH and its combination with 75, 150 and 300 mg/kg of lithium in CA1 region of hippocampus. Data are shown as means ± SD, (N = 8).* $P < 0.05$ vs. 10 mg/kg of MPH. # $P < 0.05$ vs. control group. MPH: Methylphenidate. Li; Lithium.

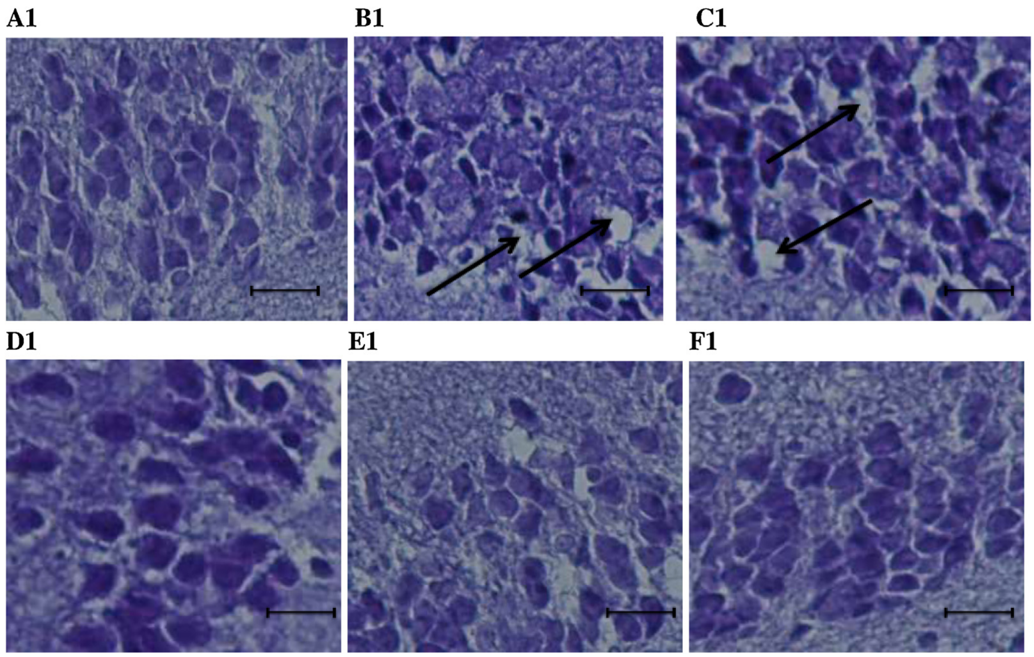

Fig. 16. Crystal violet staining shows Dentate Gyrus (DG) cell layer of the hippocampus in A1, B1, C1, D1, E1 and F1 which respectively shows group of control and under treatment with MPH (10 mg/kg) and MPH in combination with lithium (75, 150 and 300 mg/kg) and group under treatment by 150 mg/kg of lithium alone, Cell layer is different in all groups and show marked vacuolations (↑) in DG in B1 and C1 (Magnification x400. Scale bar 100 μm),(N = 8). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

findings, the harmful dose of MPH and protective doses of lithium were chosen to be studied together in order to confer the protective effects of lithium in this model. Keeping in mind the role of hippocampus in cognition, depression and anxiety like behaviors, this research was designed to look at changes that occur during these treatments in hippocampus of rats.

MPH typically used for management of ADHD in children, and is structurally and pharmacologically similar to amphetamine and

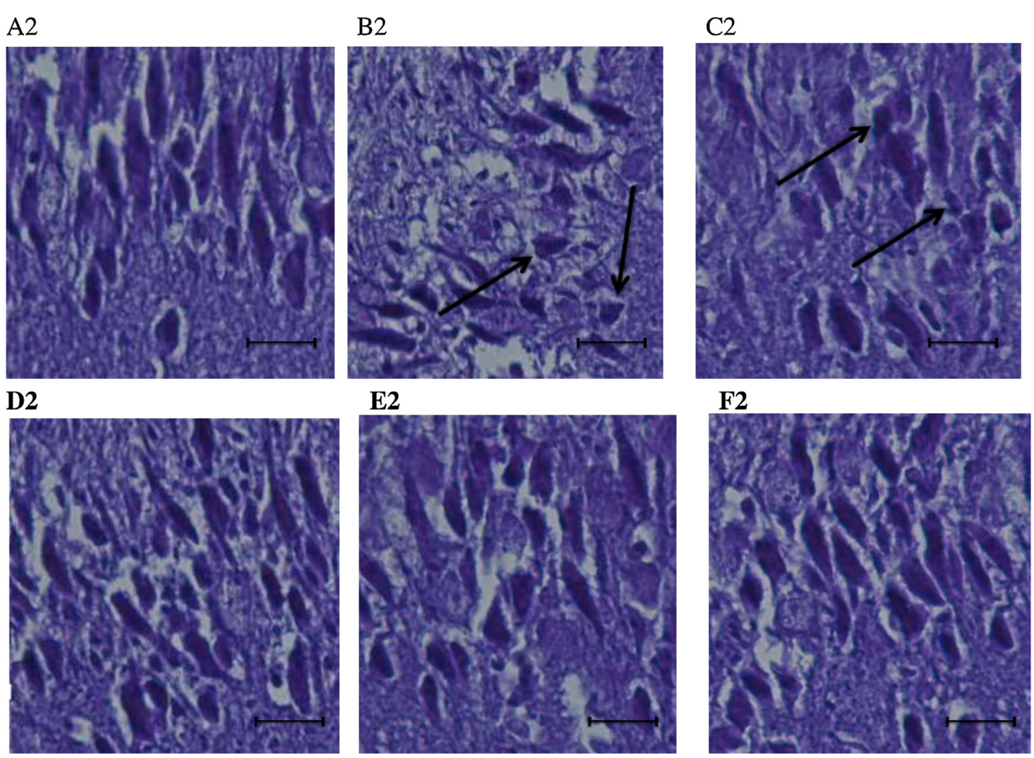

Fig. 17. Crystal violet staining shows large pyramidal cell layer of CA1 cell layer of the hippocampus in A2, B2, C2, D2, E2 and F2 which respectively shows group of control and under treatment with MPH(10 mg/kg) and MPH in combination with lithium (75, 150 and 300 mg/kg) and group under treatment by 150 mg/kg of lithium alone, Cell layer is different in all groups and show marked necrosis and shrunken (↑) in B2 and C2 (Magnification x400. Scale bar 100 μm), (N = 8). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

cocaine, and therefore has high risk of abuse (Challman and Lipsky, 2000; Motaghinejad et al., 2015c; Volkow et al., 2003). Lithium by various doses increased period of time spent in open arms (seconds) in EPM by MPH treated rats which was significant in comparison to rats treated with MPH alone (Fig. 2). Several previous studies demonstrated that lithium in sub therapeutic doses has anxiolytic and antidepressant activity (Bauer et al., 2003; Mines et al., 2010; Tondo et al., 2014). These studies showed that lithium treatment enhanced the anxiolytic-like effects of serotonergic and noradrenergic drugs and facilitate central 5-HT and NE neurotransmission at clinically therapeutic plasma lithium levels in rodents (Bauer et al., 2014; Bschor et al., 2001). Also we have shown that lithium (75, 150 and 300 mg/kg) in combination with MPH could decrease immobility and rise the swimming time in FST, while MPH (10 mg/kg) alone reduces swimming time in FST (Fig. 1). Overall, the previous findings and our results suggest that lithium in sub-therapeutic doses can improve swimming time in FST (Bersudsky et al., 2007). Also Lithium (150 mg/kg) alone increase swimming time in FST and time spent in open arms (seconds) in EPM. Using these two methods, FST and EPM, can distinguish the level of depression and anxiety in experimental groups.

Our data showed that MPH cause cognition impairment which support our pervious findings (Motaghinejad et al., 2015c,d). While in MWM test, lithium in administered doses can counteract the effects of MPH and decrease traveled distance and escaped latency in trial day and increase the percentage of time spent in target quarter in probe day in MPH dependent group (Figs. 3, 4 and 6). Treatment with Lithium (150 mg/kg) alone improved spatial learning and memory which is statistically significant when compared to MPH treated group and was not significant in comparison to control group (Vasconcellos et al., 2003; Yazlovitskaya et al., 2006). According to previous studies MPH can cause depletion of dopamine and adrenaline and this depletion could have role in cognition impairment (Motaghinejad et al., 2015c,d). While lithium increases brain adrenaline, which is important in long term potentiation and stabilization of learning and memory and lithium might inhibit the MPH induced cognition impairment (Vasconcellos et al., 2003).

MPH can cause mitochondrial dysfunction and it has been suggested that MPH can induce oxidative stress and decrease antioxidant in rat brain (Schmitz et al., 2012). MPH at a dose of 10 mg/kg causes an increase of MDA in the hippocampus while lithium in all mentioned doses can inhibit this type of lipid peroxidation (Fig. 7). This data confirmed previous study results (Andreazza et al., 2007; Schmitz et al., 2012). On the other hand, MPH (10 mg/kg) diminishes GSH and increases GSSG while lithium in mentioned doses has protective role and increased GSH and decreased GSSG content in MPH treated rats.

According to previous reports, amphetamine like compounds and MPH cause lipid peroxidation and disturb glutathione circle (Capela et al., 2007; Martins et al., 2006), While lithium has neuroprotective effects and causes decline in lipid peroxidation, and increased protective form of glutathione (GSH) and decreased its harmful and oxidized form (GSSG) (Chuang, 2004). In our study, treatment of animals with MPH (10 mg/kg) decreases GPx, antioxidant enzymes and SOD activity in isolated hippocampi. While lithium in the mentioned doses can prevent the MPH-induced decline of antioxidant enzyme activity and increases GPx and SOD activity (Fig. 8 and 9). Previous studies showed that MPH at highest therapeutic doses used can decrease SOD and GPx activities in rat hippocampus (Motaghinejad et al., 2015e). It has been proven that lithium is capable of modulating the oxidant-antioxidant activities and therefore showing the neuroprotective effects (Bhalla and Dhawan, 2009). Overall, our study showed that treatment with Lithium (150 mg/kg) alone causes decrease in MDA,GSSG and increase in GSH content and SOD and GPx activities which is statistically significant when compared to MPH only treated group and was not significant compared to control group.

Our study also showed that MPH at dose of 10 mg/kg increases the TNF-α and IL-1β levels in hippocampus. A previous study suggested that long term administration of 10 mg/kg of MPH causes increase of pro-inflammatory markers such as TNF-α and IL-1β which are is responsible for the neurodegenerative effects of MPH (Kuczenski and Segal, 2001; Motaghinejad and Motevalian, 2016; Yamamoto and Raudensky, 2008. Also our study indicated that lithium at different doses significantly decreased the inflammatory markers in MPH treated rats (Figs. 10 and 11). Consistent with our data, a previous study showed that lithium by inhibition of glycogen synthase kinase-3β enzyme (GSK-3β) and increase of BDNF expression showed protective effect against inflammation; and can attenuate TNF-α and TGF-β1 levels and decreases the inflammation and injury of kidney and liver (Jope et al., 2007; Martinez et al., 2002; Nassar and Azab, 2014). Treatment with lithium (150 mg/kg) alone causes decrease in TNF-α and IL-1β levels which is statistically significant when compared to MPH only treated group and was not significant compared to control group.

In consistency with findings of induction of oxidative stress and inflammation by MPH, our study showed that MPH (10 mg/kg) causes decrease of BDNF in both gene and protein expressions in DG and CA1 areas in hippocampus which confirmed our previous results (Motaghinejad et al., 2015f). On the other hand, MPH in combination with Lithium by mentioned doses or lithium (150 mg/kg) alone showed increase in BDNF in gene and protein levels in both DG and CA1 areas (Figs. 12–15), These results confirmed the claim of previous studies about neuroprotective properties of lithium (Fukumoto et al., 2001; Hashimoto et al., 2002). In order to show that the changes in BDNF level is related to the effect of lithium and not the behavioral tests themselves, the control groups were treated the same procedure as other groups and the only difference was administration of drugs (Angelucci et al., 2005; Li et al., 2000).

MPH (10 mg/kg) caused degeneration and death of some hippocampal cells in DG and CA1 areas. This confirms previous findings about hippocampal cell damage by methamphetamine type neurostimulants (Motaghinejad et al., 2015f; Virmani et al., 2010), while lithium (150 and 300 mg/kg) can inhibit and recover MPH induced cell death, degeneration, cell disturbance and changes in cell density, shape and integrity of hippocampus, in DG and CA1 areas (Figs. 16 and 17). This parts of data, histopathology, confirms behavioral alterations, oxidative stress parameters, inflammation biomarkers and BDNF level alteration. Also lithium alone (150 mg/kg) did not show any degeneration and alteration in shape of the cells in DG and CA1 areas in hippocampus which is significant when compared to MPH treated only group and is not significantly different compared to control group.

This neuroprotective effect of lithium might be the result of drug–drug interactions such as changing the metabolism of MPH by lithium. Although no special document was found in literature about the effect of lithium on MPH metabolism. About the effect of lithium on transporter proteins and inhibition of MPH induced neurotoxicity, it might be true, because based on previous data lithium can act on brain synaptic neurotransmitters' level and the result could be the attenuation of MPH induced neurodegeneration (Young, 2009).

Overall our data confirmed the results of previous studies about the potential neuroprotective effects of lithium in some neurodegenerative disorders in human subjects (Chuang and Priller, 2006; Martinez et al., 2002) and showed that lithium can be used in special circumstances such as abuse of methylphenidate and with possible involvement of BDNF expression can be helpful to prevent the neurodegenerative side effects of these agents.

According to our findings and the increasing rate of MPH use in human subjects, it seems that lithium is a good candidate for prevention of MPH induced neurodegeneation, anxiety, depression and cognition disturbances.

5. Conclusion

The results of the present study support the hypothesis that lithium may be beneficial against MPH induced oxidative stress, inflammation and hippocampal cell degeneration. Our data showed that chronic administration of MPH can cause increase of oxidative stress markers, inflammatory parameters, hippocampal cell loss and cause decrease of BDNF gene expression and protein production, while lithium can inhibit these effects of MPH and increases BDNF gene expression and protein production and decreases MPH induced oxidative stress, inflammation and hippocampal cell loss. Lithium can act as a neuroprotective agent against MPH induced toxicity in rat brain and this might be mediated by BDNF expression in hippocampus of rats. According to our findings and increasing rate of MPH use in human subjects, it seems that lithium is a good candidate for prevention of MPH induced neurodegeneation, anxiety, depression and cognition disturbances, although further studies are required with human subjects.

Conflict of interest

None declared.

Acknowledgement

We thank Razi Drug Research Center in Iran University of Medical Sciences for providing helpful facilities for this work.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.neuro.2016.06.010.

References

Abraham, A.D., Cunningham, C.L., Lattal, K.M., 2012. Methylphenidate enhances extinction of contextual fear. Learn. Memory 19, 67–72.
Amdisen, A., 1967. Serum lithium determinations for clinical use. Scand. J. Clin. Lab. Invest. 20, 104–108.
Andreazza, A.C., Frey, B.N., Valvassori, S.S., Zanotto, C., Gomes, K.M., Comim, C.M., et al., 2007. DNA damage in rats after treatment with methylphenidate. Prog. Neuro-Psychopharmacol. Biol. Psychiatry 31, 1282–1288.
Angelucci, F., Brene, S., Mathe, A., 2005. BDNF in schizophrenia, depression and corresponding animal models. Mol. Psychiatry 10, 345–352.
Balcioglu, A., Ren, J.-Q., McCarthy, D., Spencer, T.J., Biederman, J., Bhide, P.G., 2009. Plasma and brain concentrations of oral therapeutic doses of methylphenidate and their impact on brain monoamine content in mice. Neuropharmacology 57, 687–693.
Bauer, M., Adli, M., Baethge, C., Berghöfer, A., Sasse, J., Heinz, A., et al., 2003. Lithium augmentation therapy in refractory depression: clinical evidence and neurobiological mechanisms. Can. J. Psychiatry Revue Canadienne de Psychiatrie 48, 440–448.
Bauer, M., Bschor, T., Kunz, D., Berghöfer, A., Ströhle, A., Müller-Oerlinghausen, B., 2014. Double-blind, placebo-controlled trial of the use of lithium to augment antidepressant medication in continuation treatment of unipolar major depression. Am. J. Psychiatry 157 (9), 1429–1435.
Bedard, A.C., Tannock, R., 2007. Anxiety, methylphenidate response, and working memory in children with ADHD. J. Atten. Disord. 11 (5), 546–557.
Bersudsky, Y., Shaldubina, A., Belmaker, R., 2007. Lithium's effect in forced-swim test is blood level dependent but not dependent on weight loss. Behav. Pharmacol. 18, 77–80.
Bhalla, P., Dhawan, D., 2009. Protective role of lithium in ameliorating the aluminium-induced oxidative stress and histological changes in rat brain. Cell. Mol. Neurobiol. 29, 513–521.
Boyko, M., Nassar, A., Kaplanski, J., Zlotnik, A., Sharon-Granit, Y., Azab, A.N., 2015. Effects of acute lithium treatment on brain levels of inflammatory mediators in poststroke rats. BioMed. Res. Int. 2015.
Bruggisser, M., Bodmer, M., Liechti, M.E., 2011. Severe toxicity due to injected but not oral or nasal abuse of methylphenidate tablets. Swiss Med. Wkly. 141, w13267.
Bschor, T., Canata, B., Müller-Oerlinghausen, B., Bauer, M., 2001. Predictors of response to lithium augmentation in tricyclic antidepressant-resistant depression. J. Affect. Disord. 64, 261–265.
Capela, J.P., Fernandes, E., Remião, F., Bastos, M.L., Meisel, A., Carvalho, F., 2007. Ecstasy induces apoptosis via 5-HT 2A-receptor stimulation in cortical neurons. Neurotoxicology 28, 868–875.
Challman, T.D., Lipsky, J.J., 2000. Methylphenidate: its pharmacology and uses. Mayo Clinic Proceedings, Elsevier, pp. 711–721.
Chrislip, D., Anger, W., Smith, R., 1984. Effects of acute oral administration of lithium compounds on multiple schedule performance in rats. Neurobehav. Toxicol. Teratol. 7, 39–42.
Chuang, D.-M., Priller, J., 2006. Potential use of lithium in neurodegenerative disorders. Lithium in Neuropsychiatry: the Comprehensive Guide. Taylor & Francis Books, London, pp. 381–397.
Chuang, D.M., Chen, R.W., Chalecka-Franaszek, E., Ren, M., Hashimoto, R., Senatorov, V., et al., 2002. Neuroprotective effects of lithium in cultured cells and animal models of diseases. Bipolar Disord. 4, 129–136.
Chuang, D.-M., Wang, Z., Chiu, C.-T., 2011. GSK-3 as a target for lithium-induced neuroprotection against excitotoxicity in neuronal cultures and animal models of ischemic stroke. Front. Mol. Neurosci. 4, 15–21.
Chuang, D.-M., 2004. Neuroprotective and neurotrophic actions of the mood stabilizer lithium: can it be used to treat neurodegenerative diseases? Crit. Rev. Neurobiol. 16 (1–2), 83–90.
Dhawan, D., Singh, A., Singh, B., Bandhu, H.K., Chand, B., Singh, N., 1999. Effect of lithium augmentation on the trace elemental profile in diabetic rats. Biometals 12, 375–382.
Diniz, B.S., Machado-Vieira, R., Forlenza, O.V., 2013a. Lithium and neuroprotection: translational evidence and implications for the treatment of neuropsychiatric disorders. Neuropsychiatr. Dis. Treat. 9, 493.
Diniz, B.S., Machado-Vieira, R., Forlenza, O.V., 2013b. Lithium and neuroprotection: translational evidence and implications for the treatment of neuropsychiatric disorders. Neuropsychiatr. Dis. Treat. 9, 493–500.
Fagundes, A.O., Rezin, G.T., Zanette, F., Grandi, E., Assis, L.C., Dal-Pizzol, F., et al., 2007. Chronic administration of methylphenidate activates mitochondrial respiratory chain in brain of young rats. Int. J. Dev. Neurosci. 25, 47–51.
Faraone, S.V., Spencer, T., Aleardi, M., Pagano, C., Biederman, J., 2004. Meta-analysis of the efficacy of methylphenidate for treating adult attention-deficit/hyperactivity disorder. J. Clin. Psychopharmacol. 24, 24–29.
Frey, B.N., Valvassori, S.S., Réus, G.Z., Martins, M.R., 2006. Effects of lithium and valproate on amphetamine-induced oxidative stress generation in an animal model of mania. J. Psychiatry Neurosci.: JPN 31, 326.
Fukumoto, T., Morinobu, S., Okamoto, Y., Kagaya, A., Yamawaki, S., 2001. Chronic lithium treatment increases the expression of brain-derived neurotrophic factor in the rat brain. Psychopharmacology (Berl.) 158, 100–106.
Gerasimov, M.R., Franceschi, M., Volkow, N.D., Gifford, A., Gatley, S.J., Marsteller, D., et al., 2000. Comparison between intraperitoneal and oral methylphenidate administration: a microdialysis and locomotor activity study. J. Pharmacol. Exp. Ther. 295, 51–57.
Greenhill, L., Kollins, S., Abikoff, H., McCracken, J., Riddle, M., Swanson, J., et al., 2006. Efficacy and safety of immediate-release methylphenidate treatment for preschoolers with ADHD. J. Am. Acad. Child Adolesc. Psychiatry 45, 1284–1293.
Hajek, T., Weiner, M., 2016. Neuroprotective effects of lithium in human brain? Food for thought. Curr. Alzheimer Res. (Epub ahead of print).
Hanak, A.-S., Chevillard, L., El Balkhi, S., Risède, P., Peoc'h, K., Mégarbane, B., 2014. of blood and brain lithium pharmacokinetics in the rat according to three different modalities of poisoning. Toxicol. Sci. 143 (1), 185–195 (kfu224).
Hannestad, J., Gallezot, J.-D., Planeta-Wilson, B., Lin, S.-F., Williams, W.A., van Dyck, C.H., et al., 2010. Clinically relevant doses of methylphenidate significantly occupy norepinephrine transporters in humans in vivo. Biol. Psychiatry 68, 854–860.
Hashimoto, R., Takei, N., Shimazu, K., Christ, L., Lu, B., Chuang, D.-M., 2002. Lithium induces brain-derived neurotrophic factor and activates TrkB in rodent cortical neurons: an essential step for neuroprotection against glutamate excitotoxicity. Neuropharmacology 43, 1173–1179.
Hsieh, Y.-S., Chen, P.-N., Yu, C.-H., Chen, C.-H., Tsai, T.-T., Kuo, D.-Y., 2015. Involvement of oxidative stress in the regulation of NPY/CART-mediated appetite control in amphetamine-treated rats. Neurotoxicology 48, 131–141.
Ji, M.-H., Qiu, L.-L., Yang, J.-J., Zhang, H., Sun, X.-R., Zhu, S.-H., et al., 2015. Pre-administration of curcumin prevents neonatal sevoflurane exposure-induced neurobehavioral abnormalities in mice. Neurotoxicology 46, 155–164.
Jope, R.S., 2003. Lithium and GSK-3: one inhibitor, two inhibitory actions, multiple outcomes. Trends Pharmacol. Sci. 24, 441–443.
Jope, R.S., Yuskaitis, C.J., Beurel, E., 2007. Glycogen synthase kinase-3 (GSK3): inflammation, diseases, and therapeutics. Neurochem. Res. 32, 577–595.
Khairova, R., Pawar, R., Salvadore, G., Juruena, M.F., De Sousa, R.T., Soeiro-de-Souza, M.G., et al., 2012. Effects of lithium on oxidative stress parameters in healthy subjects. Mol. Med. Rep. 5, 680–682.
Kuczenski, R., Segal, D.S., 2001. Locomotor effects of acute and repeated threshold doses of amphetamine and methylphenidate: relative roles of dopamine and norepinephrine. J. Pharmacol. Exp. Ther. 296, 876–883.

Kuloglu, M., Ustundag, B., Atmaca, M., Canatan, H., Tezcan, A.E., Cinkilinc, N., 2002. Lipid peroxidation and antioxidant enzyme levels in patients with schizophrenia and bipolar disorder. Cell Biochem. Funct. 20, 171–175.
Lenox, R.H., Hahn, C.-G., 2000. Overview of the mechanism of action of lithium in the brain: fifty-year update. J. Clin. Psychiatry 61, 5–15.
Leutgeb, J.K., Leutgeb, S., Moser, M.-B., Moser, E.I., 2007. Pattern separation in the dentate gyrus and CA3 of the hippocampus. Science 315, 961–966.
Li, X.-C., Jarvis, E.D., Alvarez-Borda, B., Lim, D.A., Nottebohm, F., 2000. A relationship between behavior, neurotrophin expression, and new neuron survival. Proc. Natl. Acad. Sci. 97, 8584–8589.
Machado-Vieira, R., Manji, H.K., Zarate Jr, C.A., 2009. The role of lithium in the treatment of bipolar disorder: convergent evidence for neurotrophic effects as a unifying hypothesis. Bipolar Disord. 11, 92–109.
Marmol, F., 2008. Lithium: bipolar disorder and neurodegenerative diseases possible cellular mechanisms of the therapeutic effects of lithium. Prog. Neuro-Psychopharmacol. Biol. Psychiatry 32, 1761–1771.
Martinez, A., Castro, A., Dorronsoro, I., Alonso, M., 2002. Glycogen synthase kinase 3 (GSK-3) inhibitors as new promising drugs for diabetes, neurodegeneration, cancer, and inflammation. Med. Res. Rev. 22, 373–384.
Martins, M.R., Reinke, A., Petronilho, F.C., Gomes, K.M., Dal-Pizzol, F., Quevedo, J., 2006. Methylphenidate treatment induces oxidative stress in young rat brain. Brain Res. 1078, 189–197.
Mines, M.A., Yuskaitis, C.J., King, M.K., Beurel, E., Jope, R.S., 2010. GSK3 influences social preference and anxiety-related behaviors during social interaction in a mouse model of fragile X syndrome and autism. PLoS One 5 (3), e9706.
Morrison, J.M., Pritchard, H.D., Braude, M.C., D'Aguanno, W., 1971. Plasma and brain lithium levels after lithium carbonate and lithium chloride administration by different routes in rats. Exp. Biol. Med. 137, 889–892.
Motaghinejad, M., Motevalian, M., 2016. Involvement of AMPA/kainate and GABA A receptors in topiramate neuroprotective effects against methylphenidate abuse sequels involving oxidative stress and inflammation in rat isolated hippocampus. Eur. J. Pharmacol. 784, 181–191.
Motaghinejad, M., Karimian, M., Motaghinejad, O., Shabab, B., Yazdani, I., Fatima, S., 2015a. Protective effects of various dosage of curcumin against morphine induced apoptosis and oxidative stress in rat isolated hippocampus. Pharmacol. Rep. 67, 230–235.
Motaghinejad, M., Karimian, S.M., Motaghinejad, O., Shabab, B., Asadighaleni, M., Fatima, S., 2015b. The effect of various morphine weaning regimens on the sequelae of opioid tolerance involving physical dependency, anxiety and hippocampus cell neurodegeneration in rats. Fundam. Clin. Pharmacol. 29, 299–309.
Motaghinejad, M., Motevalian, M., Ebrahimzadeh, A., 2015c. Reduction of methylphenidate induced anxiety, depression and cognition impairment by various doses of venlafaxine in rat. Int. J. Preventive Med. 4 (6), 52–56.
Motaghinejad, M., Motevalian, M., Larijani, S.F., Khajehamedi, Z., 2015d. Protective effects of forced exercise against methylphenidate-induced anxiety, depression and cognition impairment in rat. Adv. Biomed. Res. 27 (4), 134–138.
Motaghinejad, M., Motevalian, M., Shabab, B., 2015e. Effects of chronic treatment with methylphenidate on oxidative stress and inflammation in hippocampus of adult rats. Neurosci. Lett. 21, 106–113.
Motaghinejad, M., Motevalian, M., Shabab, B., 2015f. Neuroprotective effects of various doses of topiramate against methylphenidate induced oxidative stress and inflammation in rat isolated hippocampus. Clin. Exp. Pharmacol. Physiol. 43 (3), 360–371.
Nassar, A., Azab, A.N., 2014. Effects of lithium on inflammation. ACS Chem. Neurosci. 5, 451–458.
Phiel, C.J., Klein, P.S., 2001. Molecular targets of lithium action. Annu. Rev. Pharmacool. Toxicol. 41, 789–813.
Quinn, D., Wigal, S., Swanson, J., Hirsch, S., Ottolini, Y., Dariani, M., et al., 2004. Comparative pharmacodynamics and plasma concentrations of d-threo-methylphenidate hydrochloride after single doses of d-threo-methylphenidate hydrochloride and d, l-threo-methylphenidate hydrochloride in a double-blind, placebo-controlled, crossover laboratory school study in children with attention-deficit/hyperactivity disorder. J. Am. Acad. Child Adolesc. Psychiatry 43, 1422–1429.
Rahimi, H.R., Dehpour, A.R., Mehr, S.E., Sharifzadeh, M., Ghahremani, M.H., Razmi, A., et al., 2014. Lithium attenuates cannabinoid-Induced dependence in the animal model: involvement of phosphorylated ERK1/2 and GSK-3 [Beta] signaling pathways. Acta Med. Iran. 52 (9), 656–663.
Rahimi_Balaei, M., Momeny, M., Babaeikelishomi, R., Mehr, S.E., Tavangar, S.M., Dehpour, A.R., 2010. The modulatory effect of lithium on doxorubicin-induced cardiotoxicity in rat. Eur. J. Pharmacol. 641, 193–198.
Riddle, E.L., Fleckenstein, A.E., Hanson, G.R., 2006. Mechanisms of methamphetamine-induced dopaminergic neurotoxicity. AAPS J. 8, E413–E418.
Sadasivan, S., Pond, B.B., Pani, A.K., Qu, C., Jiao, Y., Smeyne, R.J., 2012. Methylphenidate exposure induces dopamine neuron loss and activation of microglia in the basal ganglia of mice. PLoS One 7, e33693.
Sandoval, V., Riddle, E.L., Hanson, G.R., Fleckenstein, A.E., 2002. Methylphenidate redistributes vesicular monoamine transporter-2: role of dopamine receptors. J. Neurosci. 22, 8705–8710.
Schiffer, W., Volkow, N., Fowler, J., Alexoff, D., Logan, J., Dewey, S., 2006. Therapeutic doses of amphetamine or methylphenidate differentially increase synaptic and extracellular dopamine. Synapse 59, 243–251.
Schmitz, F., Scherer, E.B., Machado, F.R., da Cunha, A.A., Tagliari, B., Netto, C.A., et al., 2012. Methylphenidate induces lipid and protein damage in prefrontal cortex, but not in cerebellum, striatum and hippocampus of juvenile rats. Metab. Brain Dis. 27, 605–612.
Serteser, M., Ozben, T., Gumuslu, S., Balkan, S., Balkan, E., 2002. Lipid peroxidation in rat brain during focal cerebral ischemia: prevention of malondialdehyde and lipid conjugated diene production by a novel antiepileptic, lamotrigine. Neurotoxicology 23, 111–119.
Severus, E., Kleindienst, N., Seemüller, F., Frangou, S., Möller, H., Greil, W., 2007. What is the optimal serum lithium level in the long-term treatment of bipolar disorder? A review. Pharmacopsychiatry 10 (2), 231–237.
Severus, W., Kleindienst, N., Seemüller, F., Frangou, S., Möller, H., Greil, W., 2008. What is the optimal serum lithium level in the long-term treatment of bipolar disorder—a review? Bipolar Disord. 10, 231–237.
Teo, S., Stirling, D., Thomas, S., Hoberman, A., Kiorpes, A., Khetani, V., 2002. A 90-day oral gavage toxicity study of d-methylphenidate and d, l-methylphenidate in Sprague–Dawley rats. Toxicology 179, 183–196.
Thanos, P.K., Robison, L.S., Steier, J., Hwang, Y.F., Cooper, T., Swanson, J.M., et al., 2015. A pharmacokinetic model of oral methylphenidate in the rat and effects on behavior. Pharmacol. Biochem. Behav. 131, 143–153.
Tondo, L., Baldessarini, R.J., Hennen, J., Floris, G., 2014. Lithium maintenance treatment of depression and mania in bipolar I and bipolar II disorders. Am. J. Psychiatry 155 (5), 638–645.
Vasconcellos, A., Tabajara, A., Ferrari, C., Rocha, E., Dalmaz, C., 2003. Effect of chronic stress on spatial memory in rats is attenuated by lithium treatment. Physiol. Behav. 79, 143–149.
Vendruscolo, L.F., Izídio, G.S., Takahashi, R.N., Ramos, A., 2008. Chronic methylphenidate treatment during adolescence increases anxiety-related behaviors and ethanol drinking in adult spontaneously hypertensive rats. Behav. Pharmacol. 19, 21–27.
Virmani, A., Ali, S.F., Binienda, Z.K., 2010. Neuroprotective strategies in drug abuse-evoked encephalopathy. Ann. N.Y. Acad. Sci. 1199, 52–68.
Volkow, N.D., Wang, G., Fowler, J.S., Logan, J., Gerasimov, M., Maynard, L., et al., 2001. Therapeutic doses of oral methylphenidate significantly increase extracellular dopamine in the human brain. J. Neurosci. 21 (2), RC121.
Volkow, N.D., Fowler, J., Wang, G., Ding, Y., Gatley, S.J., 2003. Mechanism of action of methylphenidate: insights from PET imaging studies. J. Atten. Disord. 6, 31–44.
Wargin, W., Patrick, K., Kilts, C., Gualtieri, C., Ellington, K., Mueller, R., et al., 1983. Pharmacokinetics of methylphenidate in man, rat and monkey. J. Pharmacol. Exp. Ther. 226, 382–386.
Yamamoto, B.K., Raudensky, J., 2008. The role of oxidative stress, metabolic compromise, and inflammation in neuronal injury produced by amphetamine-related drugs of abuse. J. Neuroimmune Pharmacol. 3, 203–217.
Yazlovitskaya, E.M., Edwards, E., Thotala, D., Fu, A., Osusky, K.L., Whetsell, W.O., et al., 2006. Lithium treatment prevents neurocognitive deficit resulting from cranial irradiation. Cancer Res. 66, 11179–11186.
Young, W., 2009. Review of lithium effects on brain and blood. Cell Transplant. 18, 951–975.
Yu, F., Wang, Z., Tchantchou, F., Chiu, C.-T., Zhang, Y., Chuang, D.-M., 2012. Lithium ameliorates neurodegeneration, suppresses neuroinflammation, and improves behavioral performance in a mouse model of traumatic brain injury. J. Neurotrauma 29, 362–374.